# CONDITION W3: ABSENCE OF SIEGEL ZEROS FOR CHARACTERS MODULO $3^K$

KHAYYAM WAKIL

Abstract. We prove Condition W3: for every primitive Dirichlet character $\chi$ modulo $q = 3^K$, the $L$-function $L(s, \chi)$ has no real zero in the region $\sigma > 1 - c/\log q$, for an absolute, effectively computable constant $c > 0$. The statement is classical, and its proof takes one line: the group $(\mathbb{Z}/3^K\mathbb{Z})^\times$ is cyclic of even order, so the only real non-principal character modulo $3^K$ is the one induced from the character $\chi_{-3}$ modulo 3; an exceptional zero can only belong to a real character; and $L(\sigma, \chi_{-3}) > 0$ for $\sigma > 0$. As context we record, with corrections, the arithmetic of $\mathbb{Q}(\omega)$ that the original version used: the base change $\chi \circ \mathrm{N}$ of a character $\chi$ modulo $3^K$ takes the value $\chi(n)^2$ on the ideal $(n)$, its conductor is $\mathfrak{p}^{2K-1}$ for $K \geq 2$, and $L(s, \chi \circ \mathrm{N}) = L(s, \chi)L(s, \chi\chi_{-3})$. Condition W3 does not make any level of distribution unconditional, and nothing in it is specific to the prime 3.

**Revision note (November 2026).** The proof through a "Stark–Baker theorem" for CM fields is replaced by the one-line cyclicity argument, and the former Theorem 3.2 is withdrawn: as stated, it would solve the Landau–Siegel zero problem. The lifting identity $\chi(n) = \Psi_\chi((n))$ of the former Proposition 2.5 is withdrawn: the base change $\chi \circ \mathrm{N}$ gives $\chi(n)^2$, its conductor is $\mathfrak{p}^{2K-1}$ (not $\mathfrak{p}^{2K}$) for $K \geq 2$, and the norm map onto $(\mathbb{Z}/3^K\mathbb{Z})^\times$ has image of index 2. The claim that the argument is specific to 3 (former Proposition 5.1) is withdrawn, as is the statement that 2 splits in $\mathbb{Z}[\omega]$ (it is inert, since $-3 \equiv 5 \pmod 8$). The table of "Watt conditions" (former Theorem 7.1), the claim that W3 implies $\theta_f \leq 7/32$, the zero-density estimate quoted with it, and the "unconditional level of distribution 5/8" (former Corollary 7.2) are withdrawn. The case $K = 1$, the Minkowski bound and the computation of $\mathrm{N}(1 - \omega)$ are corrected, and the reference to a "[NOTE]" in the companion paper, which does not exist there, is removed.

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## 1. Setup and Statement

1.1. **Context.** This paper is p07 of a nine-paper programme [9]. The companion paper [10] (p04) concerns primes in arithmetic progressions to the moduli $q = 3^K$. The original version of the present paper presented Condition W3 below as the third of three hypotheses "(W1)–(W3)" of Watt's theorem [8], and claimed that proving it made a level of distribution 5/8 for these moduli unconditional. Both claims are withdrawn (Section 6): Watt's theorem has no such hypotheses, and the level 5/8 for each single modulus $3^K$ is a conjecture in [10], which would go beyond what the Generalised Riemann Hypothesis gives. The name "Condition W3" is kept as a label for the following classical statement.

1.2. **Main theorem.**

**Theorem 1.1** (Condition W3)**.** *Let $K \geq 1$ and let $\chi$ be a primitive Dirichlet character modulo $q = 3^K$. Then $L(s, \chi)$ has no real zero in the region*

$$\sigma > 1 - \frac{c}{\log q}$$

*for an absolute, effectively computable constant $c > 0$.*

The proof (Section 2) uses only the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^\times$, the classical zero-free region for Dirichlet $L$-functions, and the positivity of $L(\sigma, \chi_{-3})$. Sections 3 and 4 record, with corrections, the arithmetic of $\mathbb{Q}(\omega)$ that the original version used; it is context only and is not needed for the proof. Section 5 explains why the "Stark–Baker theorem" of the original version is withdrawn, and Section 6 explains why W3 has no consequence for any level of distribution beyond the Siegel–Walfisz range.

1.3. **Notation and conventions.** Throughout: $\omega = e^{2\pi i/3}$, so $\omega^2 + \omega + 1 = 0$. We write $\mathcal{K} = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$, with ring of integers $\mathcal{O}_\mathcal{K} = \mathbb{Z}[\omega]$. The prime $\mathfrak{p} = (1 - \omega)$ is the unique prime of $\mathbb{Z}[\omega]$ above 3, with $\mathrm{N}(\mathfrak{p}) = 3$ and $(3) = \mathfrak{p}^2$. We write $\varepsilon = \chi_{-3} = \left(\frac{-3}{\cdot}\right)$ for the non-principal character modulo 3, which is the quadratic character of $\mathcal{K}/\mathbb{Q}$. For a Dirichlet character $\chi$, $\mathfrak{f}(\chi)$ is its conductor. All constants are absolute and effectively computable unless stated otherwise.

## 2. Proof of Condition W3

**Lemma 2.1** (Real characters modulo $3^K$)**.** *For $K \geq 1$, the group $(\mathbb{Z}/3^K\mathbb{Z})^\times$ is cyclic of order $2 \cdot 3^{K-1}$, and there is exactly one real non-principal character modulo $3^K$, namely the character induced by $\chi_{-3}$. Its $L$-function is $L(s, \chi_{-3})$. In particular, for $K \geq 2$ every primitive character modulo $3^K$ is complex.*

*Proof.* The powers of 2 modulo 9 are $2, 4, 8, 7, 5, 1$, so 2 is a primitive root modulo 9, hence modulo every $3^K$ [4, Ch. 4] (and trivially modulo 3). The character group of a cyclic group of even order is cyclic of the same order and has exactly one element of order 2. The character induced by $\chi_{-3}$ is real and non-principal, so it is that element. Since $\chi_{-3}(n) = 0$ already when

$3 \mid n$, the induced character coincides with $\chi_{-3}$ as a function on $\mathbb{Z}$, and the two $L$-functions are equal. Its conductor is 3, so it is not primitive modulo $3^K$ for $K \geq 2$. □

**Lemma 2.2** (Positivity of $L(\sigma, \chi_{-3})$)**.** *$L(\sigma, \chi_{-3}) > 0$ for every real $\sigma > 0$.*

*Proof.* For $\sigma > 0$ the series $\sum_n \chi_{-3}(n) n^{-\sigma}$ converges, and grouping consecutive non-zero terms gives

$$L(\sigma, \chi_{-3}) = \sum_{m \geq 0} \big( (3m+1)^{-\sigma} - (3m+2)^{-\sigma} \big),$$

a series of positive terms. □

For example, $L(0, \chi_{-3}) = 1/3$ and $L(1, \chi_{-3}) = \pi/(3\sqrt{3}) = 0.6045\ldots$.

**Theorem 2.3** (Classical zero-free region; see [2, Ch. 14])**.** *There is an absolute, effectively computable constant $c_0 \in (0, 1]$ with the following property. Let $q \geq 3$ and let $\chi$ be a non-principal character modulo $q$. If $\chi$ is not real, $L(s, \chi)$ has no zero in the region $\sigma \geq 1 - c_0/\log\big(q(|t|+2)\big)$. If $\chi$ is real, $L(s, \chi)$ has at most one zero in this region, and such a zero is real and simple.*

*Proof of Theorem 1.1.* Let $c = c_0/2$, with $c_0$ as in Theorem 2.3, and suppose that $\beta$ is a real zero of $L(s, \chi)$ with $\beta > 1 - c/\log q$. Since $L(s, \chi) \neq 0$ for $\sigma \geq 1$, we have $\beta < 1$. Since $q \geq 3$, $\log(2q) \leq 2 \log q$, so $\beta > 1 - c_0/\log(2q)$, and $s = \beta$ lies in the region of Theorem 2.3 (with $t = 0$).

If $K \geq 2$, then $\chi$ is not real (Lemma 2.1), and Theorem 2.3 excludes $\beta$. If $K = 1$, the only primitive character modulo 3 is $\chi = \chi_{-3}$, and $L(\beta, \chi_{-3}) > 0$ by Lemma 2.2. In either case we reach a contradiction. □

*Remark* 2.4 (The one-line argument, and a slightly stronger statement). In one line: $(\mathbb{Z}/3^K\mathbb{Z})^\times$ is cyclic of even order, so the only real non-principal character modulo $3^K$ is induced from $\chi_{-3}$; an exceptional zero can only belong to a real character; so any exceptional zero modulo $3^K$ would be a zero of the single fixed function $L(s, \chi_{-3})$, which is positive on $(0, 1]$. The same argument shows that no non-principal character modulo $3^K$, primitive or not, has a zero in the region of Theorem 2.3 with $q = 3^K$: the moduli $3^K$ have no exceptional zero. Consequently the prime number theorem for arithmetic progressions to these moduli holds in the Siegel–Walfisz range with effectively computable constants [2, Ch. 20].

*Remark* 2.5 (Nothing here is specific to 3). The proof uses only that $(\mathbb{Z}/3^K\mathbb{Z})^\times$ is cyclic of even order and that one fixed real $L$-function has no real zero near 1. For every odd prime $\ell$, $(\mathbb{Z}/\ell^K\mathbb{Z})^\times$ is cyclic of even order, and the only real non-principal character modulo $\ell^K$ is induced from the Legendre symbol $(\cdot/\ell)$. Hence, for each fixed $\ell$, the same argument proves Condition W3 for the moduli $\ell^K$, with a constant $c_\ell > 0$ that depends on $\ell$ (through the largest real zero of $L(s, (\cdot/\ell))$ in $(0, 1)$, if there is one). A constant uniform in $\ell$ would exclude Siegel zeros for the real characters of prime conductor, which is an open problem. Likewise, the field $\mathbb{Q}(\omega)$ plays no special role: $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{-11})$, $\mathbb{Q}(\sqrt{-19})$, $\mathbb{Q}(\sqrt{-43})$, $\mathbb{Q}(\sqrt{-67})$ and $\mathbb{Q}(\sqrt{-163})$ also have class number 1, and each is ramified at a single prime $\ell$, with $(\ell) = \mathfrak{l}^2$.

## 3. The field $\mathbb{Q}(\omega)$ (context)

**Proposition 3.1** (Arithmetic of $\mathbb{Q}(\omega)$)**.** *The field $\mathcal{K} = \mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$ is imaginary quadratic, hence a CM field; its maximal totally real subfield is $\mathbb{Q}$. Its ring of integers is $\mathbb{Z}[\omega]$, its discriminant is $-3$, its class number is $h_{\mathcal{K}} = 1$, and its unit group is $\mu_6 = \{\pm 1, \pm\omega, \pm\omega^2\}$.*

*Proof.* Since $-3 \equiv 1 \pmod 4$, the ring of integers is $\mathbb{Z}\big[\frac{1+\sqrt{-3}}{2}\big] = \mathbb{Z}[\omega]$, with discriminant $-3$. The Minkowski bound equals $(2/\pi)\sqrt{3} \approx 1.10$, so every ideal class contains an integral ideal of norm at most 1.10, that is, of norm 1; hence $h_{\mathcal{K}} = 1$. The units are the six solutions of $a^2 - ab + b^2 = 1$, namely $\pm 1, \pm\omega, \pm\omega^2$. □

*Remark* 3.2. $\mathcal{K}$ is one of the nine imaginary quadratic fields of class number one, $\mathbb{Q}(\sqrt{-d})$ with $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$, as proved by Heegner [3], Baker [1] and Stark [7]. That $\mathbb{Z}[\omega]$ is a principal ideal domain was known long before; the class-number-one theorem plays no role in this paper.

**Proposition 3.3** (Ramification of 3)**.** *In $\mathbb{Z}[\omega]$, the rational prime 3 ramifies:* $(3) = \mathfrak{p}^2$ *where* $\mathfrak{p} = (1 - \omega)$ *and* $\mathrm{N}(\mathfrak{p}) = 3$.

*Proof.* $\mathrm{N}(1 - \omega) = (1 - \omega)(1 - \bar{\omega}) = 1 - (\omega + \bar{\omega}) + \omega\bar{\omega} = 1 - (-1) + 1 = 3$; equivalently, $\mathrm{N}(a + b\omega) = a^2 - ab + b^2$ gives $\mathrm{N}(1 - \omega) = 1 + 1 + 1 = 3$. Since $1 + \omega + \omega^2 = 0$, we have $3 = -\omega^2(1 - \omega)^2$, and $-\omega^2 \in \mathbb{Z}[\omega]^\times$. □

*Remark* 3.4. The prime 3 is the only rational prime that ramifies in $\mathcal{K}$: a prime $\ell$ ramifies in $\mathcal{K}$ if and only if $\ell \mid \mathrm{disc}(\mathcal{K}) = -3$ [6]. This is not special to 3: in the same way, $\ell$ is the only prime ramified in $\mathbb{Q}(\sqrt{-\ell})$ for $\ell = 7, 11, 19, 43, 67, 163$ (Remark 2.5).

**Lemma 3.5** (Splitting of primes $\ell \neq 3$)**.** *A prime $\ell \neq 3$ splits in $\mathcal{K}$ into two distinct primes of norm $\ell$ if $\ell \equiv 1 \pmod 3$, and is inert, with $\mathbb{Z}[\omega]/(\ell) \cong \mathbb{F}_{\ell^2}$, if $\ell \equiv 2 \pmod 3$. In particular 2 is inert and $\mathbb{Z}[\omega]/(2) \cong \mathbb{F}_4$.*

*Proof.* Since $\mathbb{Z}[\omega] \cong \mathbb{Z}[x]/(x^2 + x + 1)$, the prime $\ell$ splits if $x^2 + x + 1$ has a root modulo $\ell$ and is inert otherwise (the Dedekind–Kummer criterion [6]; for $\ell \neq 3$ there is no repeated root, as the discriminant of $x^2 + x + 1$ is $-3$). A root $r$ satisfies $r^3 = 1$ and $r \neq 1$ (as $3 \not\equiv 0 \pmod \ell$), so a root exists if and only if $\mathbb{F}_\ell^\times$ has an element of order 3, that is, if and only if $3 \mid \ell - 1$. For $\ell = 2$ this agrees with the usual criterion for 2 in $\mathbb{Q}(\sqrt{d})$ with $d \equiv 1 \pmod 4$ (2 splits if $d \equiv 1 \pmod 8$ and is inert if $d \equiv 5 \pmod 8$), since $-3 \equiv 5 \pmod 8$. □

## 4. Base change of characters modulo $3^K$ (context)

4.1. **Hecke characters of finite order.** Let $\mathfrak{m}$ be a nonzero ideal of $\mathcal{O}_{\mathcal{K}}$. Since $\mathcal{K}$ is imaginary quadratic, a Hecke character of finite order of $\mathcal{K}$ with modulus $\mathfrak{m}$ is a homomorphism $\Psi$ from the group of fractional ideals of $\mathcal{K}$ prime to $\mathfrak{m}$ to $\mathbb{C}^\times$ such that $\Psi((\alpha)) = 1$ for every $\alpha \in \mathcal{O}_{\mathcal{K}}$ with $\alpha \equiv 1 \pmod{\mathfrak{m}}$; that is, a character of the ray class group modulo $\mathfrak{m}$. Its conductor $\mathfrak{f}(\Psi)$ is the smallest such modulus. We put $\Psi(\mathfrak{a}) = 0$ for integral ideals $\mathfrak{a}$ not prime to $\mathfrak{m}$, and

$$L(s, \Psi) = \sum_{\mathfrak{a}} \Psi(\mathfrak{a})\mathrm{N}(\mathfrak{a})^{-s} = \prod_{\mathfrak{q}} \big(1 - \Psi(\mathfrak{q})\,\mathrm{N}(\mathfrak{q})^{-s}\big)^{-1}, \qquad \Re s > 1,$$

where $\mathfrak{a}$ runs over nonzero integral ideals and $\mathfrak{q}$ over prime ideals.

4.2. **The base change $\chi \circ \mathrm{N}$.**

**Proposition 4.1** (Base change of characters modulo $3^K$)**.** *Let $K \geq 1$, let $\chi$ be a Dirichlet character modulo $3^K$, and put $\Psi_\chi(\mathfrak{a}) = \chi(\mathrm{N}\mathfrak{a})$ for nonzero integral ideals $\mathfrak{a}$ of $\mathbb{Z}[\omega]$ (so $\Psi_\chi(\mathfrak{a}) = 0$ if $\mathfrak{p} \mid \mathfrak{a}$).*

*(i) $\Psi_\chi$ is a Hecke character of finite order with modulus $\mathfrak{p}^{2K}$, and $\Psi_\chi((n)) = \chi(n)^2$ for every $n \in \mathbb{Z}$.*

*(ii) The image of the norm map* $(\mathcal{O}_\mathcal{K}/\mathfrak{p}^{2K})^\times \to (\mathbb{Z}/3^K\mathbb{Z})^\times$ *is the subgroup* $\{n \equiv 1 \pmod 3\}$*, of index* 2*. Consequently* $\Psi_\chi = \Psi_{\chi\varepsilon}$*, and the map* $\chi \mapsto \Psi_\chi$ *is two-to-one.*

*(iii) If* $\chi$ *is primitive, then* $\Psi_\chi$ *is the trivial character when* $K = 1$*, and* $\mathfrak{f}(\Psi_\chi) = \mathfrak{p}^{2K-1}$ *when* $K \geq 2$*.*

*Proof.* (i) The norm is multiplicative, so $\Psi_\chi$ is multiplicative, and since units have norm 1, $\Psi_\chi((\alpha)) = \chi(\mathrm{N}\alpha)$ does not depend on the generator $\alpha$. If $\alpha \in \mathcal{O}_\mathcal{K}$ and $\alpha \equiv 1 \pmod{\mathfrak{p}^{2K}}$, then $\alpha = 1 + 3^K\beta$ with $\beta \in \mathcal{O}_\mathcal{K}$, and $\mathrm{N}(\alpha) = 1 + 3^K \operatorname{Tr}(\beta) + 3^{2K}\mathrm{N}(\beta) \equiv 1 \pmod{3^K}$, so $\Psi_\chi((\alpha)) = 1$. Finally $\mathrm{N}((n)) = n^2$.

(ii) For $\alpha = a + b\omega$ we have $\mathrm{N}(\alpha) = a^2 - ab + b^2 \equiv (a+b)^2 \pmod 3$, so every norm prime to 3 is $\equiv 1 \pmod 3$. Conversely, the squares in the cyclic group $(\mathbb{Z}/3^K\mathbb{Z})^\times$ of order $2 \cdot 3^{K-1}$ form its unique subgroup of index 2, which is therefore $\{n \equiv 1 \pmod 3\}$, and $m^2 = \mathrm{N}(m)$ for $m \in \mathbb{Z}$. Hence the image is exactly $\{n \equiv 1 \pmod 3\}$. Two characters modulo $3^K$ agree on this subgroup if and only if they are equal or differ by the unique non-trivial character trivial on it, which is $\varepsilon$; this gives the last two assertions.

(iii) For $K = 1$, $\chi = \varepsilon$, and $\varepsilon(\mathrm{N}\mathfrak{a}) = 1$ for every $\mathfrak{a}$ prime to $\mathfrak{p}$, by (ii). Let $K \geq 2$. One checks that $\operatorname{Tr}(\mathcal{O}_\mathcal{K}) = \mathbb{Z}$ and $\operatorname{Tr}(\mathfrak{p}) = 3\mathbb{Z}$ (indeed $\operatorname{Tr}((1-\omega)(a+b\omega)) = 3a$); since $\mathfrak{p}^{2j} = 3^j\mathcal{O}_\mathcal{K}$, it follows that $\operatorname{Tr}(\mathfrak{p}^m) = 3^{\lceil m/2\rceil}\mathbb{Z}$. If $\alpha = 1 + \beta$ with $\beta \in \mathfrak{p}^{2K-1}$, then $\mathrm{N}(\alpha) = 1 + \operatorname{Tr}(\beta) + \mathrm{N}(\beta) \equiv 1 \pmod{3^K}$, because $\operatorname{Tr}(\beta) \in 3^K\mathbb{Z}$ and $\mathrm{N}(\beta) \in 3^{2K-1}\mathbb{Z}$. So $\mathfrak{p}^{2K-1}$ is a modulus for $\Psi_\chi$. On the other hand, for $t \in \mathbb{Z}$ the element $\alpha = 1 + 3^{K-1}t$ is $\equiv 1 \pmod{\mathfrak{p}^{2K-2}}$, and $\mathrm{N}(\alpha) = (1 + 3^{K-1}t)^2 \equiv 1 + 2 \cdot 3^{K-1}t \pmod{3^K}$ because $2K - 2 \geq K$; these values run through all classes $\equiv 1 \pmod{3^{K-1}}$. A primitive $\chi$ is non-trivial on that subgroup, so $\mathfrak{p}^{2K-2}$ is not a modulus for $\Psi_\chi$, and $\mathfrak{f}(\Psi_\chi) = \mathfrak{p}^{2K-1}$. □

The conductor in (iii) agrees with the conductor–discriminant formula $|\mathrm{disc}(\mathcal{K})| \cdot \mathrm{N}\mathfrak{f}(\Psi_\chi) = \mathfrak{f}(\chi)\,\mathfrak{f}(\chi\varepsilon)$, which for $K \geq 2$ reads $3 \cdot 3^{2K-1} = 3^K \cdot 3^K$.

*Remark* 4.2 (The lifting identity of the original version). The original version (former Proposition 2.5) claimed, for every primitive $\chi$ modulo $3^K$, a primitive Hecke character $\Psi_\chi$ of conductor $\mathfrak{p}^{2K}$ with $\chi(n) = \Psi_\chi((n))$ for all $n$ prime to 3. No such character exists when $\chi$ is odd: since $(n) = (-n)$, any function of ideals takes the same value at $n$ and $-n$, which forces $\chi(-1) = 1$. Half of the characters modulo $3^K$ are odd, because $-1$ is the element of order 2 of the cyclic group $(\mathbb{Z}/3^K\mathbb{Z})^\times$. The construction actually given was $\Psi_\chi = \chi \circ \mathrm{N}$, for which (i) gives $\chi(n)^2$; its conductor is $\mathfrak{p}^{2K-1}$, not $\mathfrak{p}^{2K}$, by (iii); and the norm map is not surjective, by (ii). In particular the ramification of 3 does not make conductors "match". The opposite is true: for a prime $\ell \neq 3$, which is unramified, the base change of a primitive character modulo $\ell^K$ has conductor exactly $\ell^K\mathcal{O}_\mathcal{K}$, whereas at the ramified prime 3 the conductor drops to $\mathfrak{p}^{2K-1}$.

### 4.3. **The $L$-function factorisation.**

**Lemma 4.3** ($L$-function factorisation)**.** *For every Dirichlet character* $\chi$ *modulo* $3^K$*,* $K \geq 1$*,*

$$L(s, \Psi_\chi) = L(s, \chi) \cdot L(s, \chi\varepsilon) \qquad (\Re s > 1),$$

*and hence for all* $s$*, by analytic continuation.*

*Proof.* Both sides are Euler products; we compare the local factors at each rational prime $\ell$.

$\ell = 3$. $\Psi_\chi(\mathfrak{p}) = \chi(3) = 0$, and $\chi(3) = \chi\varepsilon(3) = 0$, so both local factors are 1.

$\ell \equiv 1 \pmod 3$ *(split).* Write $\ell\mathbb{Z}[\omega] = \mathfrak{q}_1\mathfrak{q}_2$ with $\mathfrak{q}_1 \neq \mathfrak{q}_2$ of norm $\ell$. The local factor of $L(s, \Psi_\chi)$ is $(1 - \chi(\ell)\ell^{-s})^{-2}$. Since $\varepsilon(\ell) = 1$, this equals the product of the local factors of $L(s, \chi)$ and $L(s, \chi\varepsilon)$.

$\ell \equiv 2 \pmod 3$ *(inert, including $\ell = 2$).* Then $\ell\mathbb{Z}[\omega] = \mathfrak{q}$ is a prime of norm $\ell^2$, and $\Psi_\chi(\mathfrak{q}) = \chi(\ell^2) = \chi(\ell)^2$. Since $\varepsilon(\ell) = -1$, the product of the local factors of $L(s, \chi)$ and $L(s, \chi\varepsilon)$ is $(1-\chi(\ell)\ell^{-s})^{-1}(1+\chi(\ell)\ell^{-s})^{-1} = (1-\chi(\ell)^2\ell^{-2s})^{-1}$, the local factor of $L(s, \Psi_\chi)$. □

*Remark* 4.4 (The twist $\chi\varepsilon$, and the case $K = 1$). For $K \geq 2$ and $\chi$ primitive, $\chi\varepsilon$ is again primitive modulo $3^K$, since $\varepsilon$ is trivial on $\{n \equiv 1 \pmod{3^{K-1}}\}$, and $\Psi_{\chi\varepsilon} = \Psi_\chi$. For $K = 1$, $\chi = \varepsilon$, and $\chi\varepsilon = \varepsilon^2$ is the principal character modulo 3, of conductor 1. Then $\Psi_\varepsilon$ is trivial, and Lemma 4.3 reads $\zeta_{\mathcal{K}}(s)(1 - 3^{-s}) = L(s, \varepsilon)\,\zeta(s)(1 - 3^{-s})$, that is, $\zeta_{\mathcal{K}}(s) = \zeta(s)L(s, \varepsilon)$. The original version stated that $\chi\varepsilon$ is non-principal of conductor $3^K$ for all $K \geq 1$, and that $\Psi_\varepsilon$ is primitive of conductor $\mathfrak{p}^2$; both are false for $K = 1$. Indeed the ray class group of $\mathcal{K}$ modulo $\mathfrak{p}^2$ is trivial, since the six units map onto $(\mathcal{O}_{\mathcal{K}}/\mathfrak{p}^2)^\times$, which has order 6; so no non-trivial character of conductor $\mathfrak{p}^2$ exists.

*Remark* 4.5. Lemma 4.3 expresses $L(s, \Psi_\chi)$ through two Dirichlet $L$-functions modulo $3^K$. A zero-free region for $L(s, \Psi_\chi)$ is therefore the same statement as a zero-free region for the pair $L(s, \chi)$, $L(s, \chi\varepsilon)$, and the passage to $\mathbb{Q}(\omega)$ adds nothing to Section 2.

## 5. The former "Stark–Baker theorem" (withdrawn)

The original version derived Theorem 1.1 from the following statement, presented as a "Stark–Baker theorem" (former Theorem 3.2): *for every CM field $F$ and every Hecke character $\Psi$ of $F$ of conductor $\mathfrak{f}$, $L(s, \Psi)$ has no real zero in $\sigma > 1 - c_F/\log(\mathrm{N}(\mathfrak{f})\,|\mathrm{disc}(F)|)$, with $c_F > 0$ effective and depending only on $[F : \mathbb{Q}]$.* This is not a known theorem, and it is withdrawn, together with its proof sketch and the former Remark 3.3 and Corollary 3.4.

As stated it would solve the Landau–Siegel zero problem. Take $F = \mathbb{Q}(\sqrt{-d})$, any imaginary quadratic field (hence a CM field of degree 2), and $\Psi$ trivial. Then $L(s, \Psi) = \zeta_F(s) = \zeta(s)L(s, \chi_{d_F})$, where $d_F$ is the discriminant of $F$ and $\chi_{d_F}$ the associated quadratic character, and $\zeta(\sigma) < 0$ for $0 < \sigma < 1$. The statement would therefore give that $L(s, \chi_{d_F})$ has no real zero in $\sigma > 1 - c/\log|d_F|$ for every imaginary quadratic field, with $c$ absolute and effective. That is the open Landau–Siegel problem for these characters; via the class number formula it would also give an effective lower bound $h(d_F) \gg |d_F|^{1/2}/\log|d_F|$, far beyond the known effective lower bounds.

What Heegner [3], Baker [1] and Stark [7] proved is the class-number-one theorem: the nine fields listed after Proposition 3.1 are the only imaginary quadratic fields of class number 1. It gives no zero-free region for Hecke $L$-functions of $\mathbb{Q}(\omega)$. The remark that Siegel zeros "remain possible" only for non-CM fields is also withdrawn, since every imaginary quadratic field is a CM field. None of this affects Theorem 1.1, whose proof in Section 2 uses only the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^\times$.

## 6. Relation to the companion paper and to Watt's theorem

The original Sections 5–7 (a proposition that the argument is specific to 3, a "coherence" proposition, a table verifying "Watt's conditions" W1–W3, and a corollary asserting that the level of distribution $\theta_W = 5/8$ for cascade moduli holds unconditionally) are withdrawn, for the following reasons.

(1) *Watt's theorem has no conditions W1–W3.* N. Watt [8] proves a mean-value estimate in the $t$-aspect for $\int_0^T |\zeta(\frac12+it)|^4 \left|\sum_{m\le M} a_m m^{-it}\right|^2 dt$. It involves no family of Dirichlet characters modulo $q$ and has no hypotheses "$M \le q$", "$MR \le q^2$" or "no Siegel zeros". Moreover, with the parameters $M = Q^{2/3}$ and $R = Q^{1/3}$ used in the original version of [10], "W1" fails for every $3^K < Q^{2/3}$ and "W2" for every $3^K < Q^{1/2}$, so these were not verified for the family either (former Theorem 7.1).

(2) *W3 is unrelated to $\theta \le 7/32$.* The bound $\theta \le 7/32$ is the Kim–Sarnak bound [5] $\lambda_1 \ge 1/4 - (7/64)^2$ towards Selberg's eigenvalue conjecture, written in the normalisation $\theta = \sqrt{1-4\lambda_1}$. It is an unconditional theorem about Maass forms of every level, and neither the absence of Siegel zeros nor any zero-density estimate implies it. The derivation of it in the original version is withdrawn, together with the zero-density estimate quoted there, whose exponent and source we could not confirm.

(3) *W3 makes nothing unconditional.* Theorem 1.1 of [10], a sum over the moduli $3^K \le x^\theta$, holds for every fixed $\theta < 1$ by the trivial bound and the Siegel–Walfisz theorem. It does not need W3, although Remark 2.4 makes its implied constant effective. The per-modulus statement for $3^K$ up to $x^{5/8-\varepsilon}$ is a conjecture in [10], which would go beyond the Generalised Riemann Hypothesis. A zero-free region of width $c/\log q$ cannot help with it: for $q = x^\theta$ one has $x^{1-c/\log q} = e^{-c/\theta}x$, so such a region gives no saving for moduli of size a power of $x$. The former Corollary 7.2 is therefore withdrawn. (Its reference to an annotation "[NOTE]" in [10] was also wrong: there is none.)

(4) *No "coherence".* The former Proposition 6.1 asserted that one fact, the ramification $(3) = \mathfrak{p}^2$, drives three steps of [10]: a three-level filtration, Kloosterman cancellation beyond the Weil bound, and a conductor match. All three are false. The $(1-\omega)$-adic filtration of $\mathbb{Z}[\omega]/(3^K)$ has $2K$ steps, not three. The bound $|S(a,b;3^K)| \ll (3^K)^{3/8+\varepsilon}$ fails, since $S(1,1;3^K) = 2\cdot 3^{K/2}\cos(4\pi/3^K)$ for even $K \ge 2$ (see [10]); and cyclicity of $(\mathbb{Z}/\ell^K\mathbb{Z})^\times$ holds for every odd prime $\ell$ and has nothing to do with $\mathbb{Z}[\omega]$. The conductor does not match (Remark 4.2).

(5) *Not specific to 3.* The former Proposition 5.1 is withdrawn; see Remark 2.5. Its proof also stated that 2 splits in $\mathcal{K}$ "since $-3 \equiv 1 \pmod 8$". In fact $-3 \equiv 5 \pmod 8$, and 2 is inert (Lemma 3.5). Its third property, that characters of $(\mathbb{Z}/3^K\mathbb{Z})^\times$ correspond precisely to Hecke characters of $\mathcal{K}$ of conductor $\mathfrak{p}^{2K}$, is false as well: the map $\chi \mapsto \chi\circ\mathrm{N}$ is two-to-one (Proposition 4.1(ii)), and the ray class group of $\mathcal{K}$ modulo $\mathfrak{p}^{2K}$ has order $3^{2K-2}$, not $2\cdot 3^{K-1}$.

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The ARC Institute of Knowware, Calgary, Alberta, Canada
*Email address*: the@knowware.institute